

Randomized and low-rank approximation

Query complexity from Schatten to spectral norms

Difficulty: extreme **Importance:** interesting to the community

Problem statement

For $1 \leq p < \infty$, define the Schatten norm of a real matrix M by

$$\|M\|_{\mathcal{S}_p} = \left(\sum_i \sigma_i(M)^p \right)^{1/p},$$

where $\sigma_i(M)$ are its singular values; set $\|M\|_{\mathcal{S}_\infty} = \max_i \sigma_i(M)$.

Let $n \geq 2$, $p \in [1, \infty]$, and $0 < \varepsilon < 1/2$. A randomized algorithm accesses an arbitrary unknown $A \in \mathbb{R}^{n \times n}$ only through exact products Ax or $A^T x$. Each product costs one query. Query vectors may depend on previous answers and internal randomness; computation between queries is unrestricted.

Define $q_p(n, \varepsilon)$ to be the least integer q such that some algorithm using at most q queries outputs a real unit vector v with

$$\Pr \left[\|A(I - vv^T)\|_{\mathcal{S}_p} \leq (1 + \varepsilon) \min_{\|u\|_2=1} \|A(I - uu^T)\|_{\mathcal{S}_p} \right] \geq \frac{99}{100}$$

for every A .

Open problem. Determine $q_p(n, \varepsilon)$ up to universal constant factors simultaneously in p , n , and ε , including the transition when finite p grows with n or ε^{-1} and the endpoint $p = \infty$. Finite-dimensional saturation must be included, since n column queries reveal A exactly.

This square-matrix formulation editorially fixes a two-sided oracle: the source introduction describes Av , but Algorithm 7.4 uses A and A^T . These models coincide on its symmetric lower-bound inputs. The relative error is in the norm, not its p th power.

Why it matters in numerical linear algebra

This asks how the matrix-product cost changes between aggregate singular-value error and worst-direction error.

References

1. Ainesh Bakshi and Shyam Narayanan, *Krylov Methods are (nearly) Optimal for Low-Rank Approximation*, arXiv:2304.03191v1 (2023), Open Question 1.11; Theorems 1.1 and 1.5, Algorithm 7.4.
2. Praneeth Kacham and David P. Woodruff, *Faster Algorithms for Schatten- p Low Rank Approximation*, APPROX/RANDOM 2024, article 55: later running-time bounds.

Status check

For fixed finite p , the source gives a dimension-independent query bound; the spectral lower bound grows with $\log n$ in its stated dimension regime. Neither gives optimal uniform dependence on growing p . The 2024 work improves running time.

Checked 2026-09-08: both arXiv records list only v1. Searches for “Open Question 1.11,” Schatten phase transition, query complexity, and 2025/2026 found no resolution. The older rank-one spectral lower-bound question was solved by the 2023 source. This is a bounded check.